ELEG 5693 Project 4
OFDM

I. Objectives

1. Understand the concepts of OFDM.
2. Learn to design the transmitter and receiver of OFDM.
3. Learn to simulate quasi-static frequency selective fading channels.

II. Requirements

1. Perform the simulation of a wireless communication system that experience quasi-static frequency selective fading and additive white Gaussian noise.
    a) Quasi-static fading: the channel keeps constant in one block, and changes from block to block. The channel in different blocks are independent.
    b) Fading distribution: Rayleigh fading
    c) Channel power delay profile: [0.25, 0.25, 0.25, 0.25].
    d) Modulation: QPSK
    e) Channel coding: none.
    f) Number of sub-carriers: N = 64
    g) Number of cyclic prefix: u = 6
2. Show the BER v.s. Eb/N0. The range of Eb/N0: [0:4:24] dB